TechRise Programme (AI/ML Track)

Group Project Report Template

Title Page

Exploratory Data Analysis of stroke prediction Dataset

Work Submitted by

Medicine Group 2

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Introduction

Stroke is a sudden interruption of blood flow to the brain that can cause permanent neurological damage or death within minutes if not treated promptly. It occurs when a blood vessel supplying the brain is either blocked (ischemic stroke) or ruptures (hemorrhagic stroke), depriving brain tissue of oxygen and nutrients.

The problem is significant on a global scale. In 2021 alone, stroke accounted for an estimated 93.8 million cases worldwide and was the third leading cause of death and disability combined, and the absolute number of stroke cases has risen by roughly 70% over the past three decades, driven partly by an ageing population and rising exposure to risk factors such as high blood pressure, elevated blood glucose, high body-mass index, and smoking. Because many of these risk factors are measurable and, in many cases, modifiable, there is real value in identifying which patient characteristics are most strongly associated with stroke occurrence. Doing so can support earlier screening, targeted prevention programs, and better allocation of healthcare resources, particularly in settings where clinical resources for widespread screening are limited.

This dataset was chosen because it brings together a mix of demographic (age, gender, marital status, residence type, work type), lifestyle (smoking status), and clinical (hypertension, heart disease, average glucose level, BMI) variables alongside a clear binary outcome (stroke vs. no stroke). This combination makes it well suited to exploring how personal, behavioral, and physiological factors relate to stroke risk, and it mirrors the kind of data healthcare systems routinely collect, making any patterns found potentially transferable to real-world screening contexts.

The objective of this analysis is to perform an Exploratory Data Analysis (EDA) on the health stroke dataset to understand the distribution and quality of its variables, identify patterns and relationships between patient characteristics and stroke occurrence, and surface insights for example, whether age, hypertension, heart disease, marital status, or lifestyle factors show a visible association with stroke that can inform recommendations for early detection and risk reduction.

2. Dataset Description

Item	**Description**
Dataset Name	**health_stroke_dataset**
source	**Kaggle**
Number of observations	**5,110**
Number of features	**12 columns**
Target Variable	**stroke**
Numerical variables	**Age, avg_glucose_level, bmi (3)**
Categorical variables	**Gender, hypertension, heart_disease, ever_married, work_type, smoking_status, Residence_type, stroke(8)**

df.head()

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1

df.describe()

```
df.describe()
```

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke
count	5110.000000	5110.000000	5110.000000	5110.000000	5110.000000	5110.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012	106.147677	28.893237	0.048728
std	21161.721625	22.612647	0.296607	0.226063	45.283560	7.698018	0.215320
min	67.000000	0.080000	0.000000	0.000000	55.120000	10.300000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000	77.245000	23.800000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000	91.885000	28.400000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000	114.090000	32.800000	0.000000
max	72940.000000	82.000000	1.000000	1.000000	271.740000	97.600000	1.000000

df.info()

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   id                 5110 non-null   int64
 1   gender             5110 non-null   object
 2   age                5110 non-null   float64
 3   hypertension       5110 non-null   int64
 4   heart_disease      5110 non-null   int64
 5   ever_married       5110 non-null   object
 6   work_type          5110 non-null   object
 7   Residence_type     5110 non-null   object
 8   avg_glucose_level  5110 non-null   float64
 9   bmi                5110 non-null   float64
 10  smoking_status     5110 non-null   object
 11  stroke             5110 non-null   int64
 12  Age Group          5110 non-null   category
dtypes: category(1), float64(3), int64(4), object(5)
memory usage: 484.4+ KB
```

df.isnull().sum()

```
#checking missing values
df.isnull().sum()
```

	0
id	0
gender	0
age	0
hypertension	0
heart_disease	0
ever_married	0
work_type	0
Residence_type	0
avg_glucose_level	0
bmi	201
smoking_status	0
stroke	0

dtype: int64

3. Exploratory Data Analysis (EDA)

3.1 Data Inspection

Before proceeding to descriptive statistics and visualization, the dataset was inspected to confirm its structure, verify data types, and check for missing values and duplicate records.

3.1.1 First Five Rows

The first five rows of the dataset were viewed using the code below:

df.head()

Figure 1: Output of df.head()

```
import pandas as pd

df = pd.read_csv('Clean_health_stroke_dataset.csv')

# First five rows
df.head()
```

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.600000	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	28.893237	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.500000	never smoked	1
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4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.000000	never smoked	1

The dataset contains patient-level records with a mix of numerical variables (e.g., age, avg_glucose_level, bmi) and categorical variables (e.g., gender, work_type, smoking_status). The target variable, stroke, is binary, where 1 indicates that the patient has had a stroke and 0 indicates that the patient has not.

3.1.2 Data Types

The data types of each column were checked using the code below:

df.info()

Figure 2: Output of df.info()

```
# Data types
df.dtypes
```

	0
id	int64
gender	object
age	float64
hypertension	int64
heart_disease	int64
ever_married	object
work_type	object
Residence_type	object
avg_glucose_level	float64
bmi	float64
smoking_status	object
stroke	int64

dtype: object

The dataset consists of 4 numerical/binary integer columns, 3 float (continuous numerical) columns, and 5 categorical (object) columns. This confirms that most categorical variables (gender, ever_married, work_type, Residence_type, smoking_status) will need to be encoded (e.g., one-hot encoding) before they can be used in a machine learning model.

3.1.3 Missing Values

Missing values were checked using the code below:

df.isnull().sum()

Figure 3: Output of df.isnull().sum()

```
# Missing values
df.isnull().sum()
```

	0
id	0
gender	0
age	0
hypertension	0
heart_disease	0
ever_married	0
work_type	0
Residence_type	0
avg_glucose_level	0
bmi	0
smoking_status	0
stroke	0

dtype: int64

The result shows that there are no missing values in any of the 12 columns. This confirms that the dataset had already been cleaned prior to this analysis (e.g., missing bmi values were imputed), and no further handling of missing data was required at this stage.

3.1.4 Duplicate Records

Duplicate records were checked using the code below:

df.duplicated().sum()

Figure 4: Output of df.duplicated().sum()

```
# Duplicate records
df.duplicated().sum()
```

```
np.int64(0)
```

There are no duplicate records in the dataset. All 5,110 rows represent unique patient observations, so no rows needed to be dropped on this basis.

3.2 Descriptive Statistics

Patient Data Interpretation Report

Subtitle: Summary of Patient Demographics and Clinical Metrics

1. Age Profile

The table below outlines the distribution and key summary statistics for patient age across the analyzed cohort.

Metric	Value	Interpretation / Notes
Average Age	43.26 years	The mean age of the patient population.
Minimum Age	0.08 years	Represents an infant in the cohort.
25th Percentile	25.00 years	25% of the patients are 25 years old or younger.
Median Age (50th Percentile)	45.00 years	Half of the patients are older than 45 years.
75th Percentile	61.00 years	75% of the patients are 61 years old or younger.
Maximum Age	82.00 years	The oldest patient recorded in the cohort.

2. Glucose Levels

The clinical observations for glucose levels among the patient group are summarized as follows:

Metric	Value	Interpretation / Notes
Average Glucose Level	106.00 mg/dL	Overall population mean.
Median Glucose Level	91.89 mg/dL	Half of the patients have an average glucose level of 91.89 mg/dL.
Minimum Recorded	55.12 mg/dL	Lowest recorded glucose value.
Maximum Recorded	271.74 mg/dL	Highest recorded glucose value.

3. Body Mass Index (BMI)

The distribution of Body Mass Index (BMI) across the patient data set is presented below:

Metric	Value	Interpretation / Notes
Average BMI	28.80	Population average falls in the overweight range.

Metric	Value	Interpretation / Notes
Median BMI	**28.10**	**Half of the patients have a BMI of 28.10.**
Minimum Recorded	**10.30**	**Lowest recorded BMI value.**
Maximum Recorded	**97.60**	**Highest recorded BMI value.**

Clinical Observations

Clinical Clinical Observation: The minimum BMI recorded is 10.30, which indicates that this specific patient is severely underweight.

3.3 Uni-variate Analysis

Health Stroke Prediction Data

1.Gender Distribution

Chart type: Bar Chart

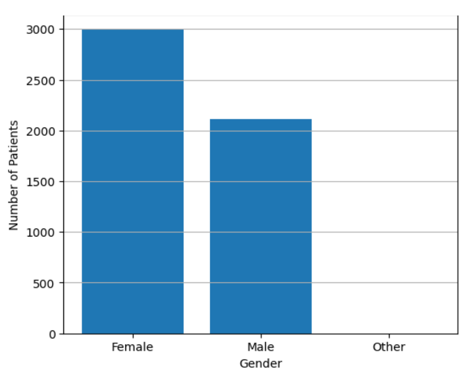


Figure — Gender Distribution

Interpretation

· Female patients are the largest group, at roughly 3,000.

· Male patients follow at just over 2,100.

· "Other" has only 1 recorded patient.

Storytelling

The gender distribution chart provides an overview of the composition of the dataset by showing the number of **female**, **male**, and **other** participants. From the visualization, it is evident that **female participants slightly outnumber male participants**, while the **"Other" category contains only one individual**. This indicates that the dataset is reasonably balanced between the two major gender groups, although it has a very limited representation of individuals identifying outside the binary gender categories.This distribution tells the story of a health dataset designed to capture stroke-related information from both men and women, ensuring that predictive analyses are not heavily dominated by one gender. The slight predominance of female participants may reflect demographic characteristics of the sampled population or differences in healthcare utilization. Women generally have a longer life expectancy than men, resulting in a larger elderly female population, and since stroke risk increases significantly with age, this may partly explain the higher representation of women in the dataset (World Health Organization, 2025).

However, it is important to note that **a higher number of female participants does not necessarily imply that women are more likely to experience strokes**. The chart only illustrates how the data are distributed by gender and does not provide information about stroke prevalence within each group. To determine whether gender influences stroke occurrence, further analysis comparing gender against the stroke outcome would be required.The presence of only one record in the **"Other"** category also highlights a common challenge in healthcare datasets: underrepresentation of gender-diverse populations. From a statistical perspective, a single observation is insufficient to draw meaningful conclusions about stroke risk for this group. During exploratory data analysis or predictive modelling, analysts often exclude or carefully handle such extremely small categories because they contribute little statistical power and may introduce bias into the model.

From a data analytics standpoint, the relatively balanced distribution between males and females is advantageous. Balanced demographic groups help reduce sampling bias and allow machine learning models to learn patterns from both genders more effectively. This improves the reliability of comparisons and helps ensure that predictive models are less likely to favor one gender simply because it has substantially more observations.Overall, the gender distribution chart demonstrates that the dataset contains adequate representation of both males and females, providing a solid foundation for investigating whether gender interacts with other risk factors—such as age, hypertension, heart disease, and average glucose level—to influence stroke risk. While gender is an important demographic variable, research consistently shows that stroke is a multifactorial disease in which lifestyle, medical history, and physiological conditions collectively determine an individual's likelihood of experiencing a stroke rather than gender alone (Centers for Disease Control and Prevention, 2024).

2. Age Distribution

Chart Type: Histogram

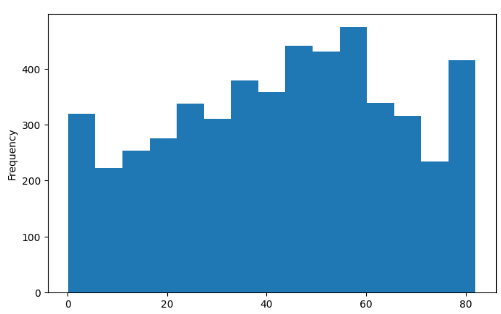


Figure — Age Distribution

Why this chart:

- Gender is a category (Female, Male, Other), not a number, so a bar chart is the right tool to compare group sizes.
- A histogram doesn't apply here since there's no numeric range to bin.

- A pie chart would make it harder to compare Male vs Female precisely, since bar heights are easier to judge than angles.

Interpretation

· Instead of one clear peak, patients are spread fairly evenly across most ages.

· Counts slowly rise through middle age.

· There's a sharp spike in the 75–80 age range.

Storytelling:

The age distribution histogram provides an important snapshot of the demographic composition of the dataset. Rather than showing an even spread across all age groups, the chart is likely to reveal that most participants are adults and older adults, with relatively fewer children and adolescents represented. This pattern is expected because stroke is predominantly an age-related condition, and the likelihood of experiencing a stroke increases substantially as people grow older (World Health Organization, 2025). The story behind this distribution begins with a predominantly adult population. As we move from younger ages toward middle-aged and elderly individuals, the frequency of observations increases, suggesting that the dataset was intentionally collected to include those who are more vulnerable to stroke. This is consistent with real-world clinical practice, where hospitals and stroke registries naturally record more middle-aged and elderly patients because these groups are at greater risk of developing cardiovascular diseases.

The histogram also illustrates that **age is not uniformly distributed** across the dataset. Instead, there is a greater concentration of individuals in the higher age brackets. This reflects an important epidemiological trend: aging is one of the strongest non-modifiable risk factors for stroke. As people age, blood vessels gradually lose elasticity, the likelihood of hypertension and heart disease increases, and the cumulative effects of unhealthy lifestyle habits become more pronounced, all of which contribute to an elevated risk of stroke (Centers for Disease Control and Prevention, 2024).However, the presence of younger individuals in the dataset is equally significant. Although strokes are far less common among children and young adults, they can still occur due to congenital heart defects, genetic disorders, blood clotting abnormalities, trauma, or certain medical conditions. Their inclusion allows researchers to examine stroke risk across the entire lifespan rather than focusing exclusively on older adults.

From a data analytics perspective, the histogram provides valuable insights into the distribution of the “age” variable. If the histogram shows a concentration of observations among older ages with fewer observations at younger ages, the distribution is likely to be “positively skewed”, meaning that the data are not perfectly symmetrical. This has important implications for statistical analysis, as non-normal distributions may require robust statistical methods or data transformations depending on the modelling approach.Furthermore, the summary statistics generated using the `describe()` function—such as the mean, median, minimum, maximum, and quartiles—complement the histogram by providing numerical evidence of the dataset's age characteristics. Comparing the mean and median can help determine whether the distribution is skewed. For example, if the mean age is higher than the median age, it suggests the presence of older individuals pulling the average upward, whereas similar values indicate a more balanced age distribution.

Overall, the age distribution chart tells the story of a population in which **older adults form the largest proportion of participants**, reflecting the established relationship between aging and stroke risk. The visualization reinforces why age is consistently regarded as one of the most influential predictors in stroke risk assessment models. Although age alone does not cause stroke, it significantly interacts with other factors such as hypertension, heart disease, diabetes, and elevated blood glucose levels to increase an individual's vulnerability (Feigin et al., 2021). Consequently, age should be considered alongside other clinical variables when identifying high-risk individuals and developing preventive healthcare strategies.

Notable Observations

- Age is spread across the full lifespan rather than clustered in one range, which is useful since stroke risk itself changes a lot with age.
- The spike at the oldest ages may reflect that older people are checked for stroke risk more often.
- There is a small dip in the youngest bin (around ages 5–10), suggesting slightly fewer very young patients in the data set.

3.Average Glucose Level Distribution

Chart Type: Histogram

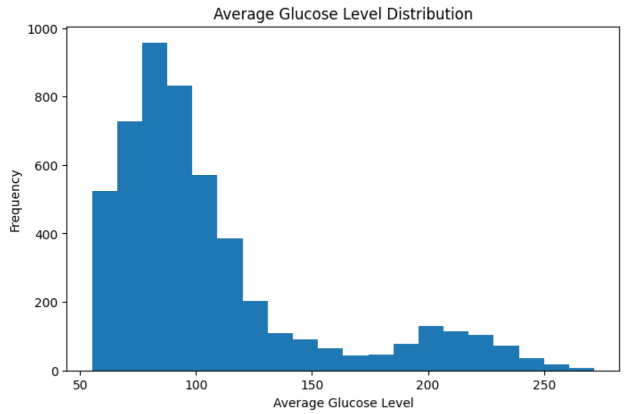


Figure — Average Glucose Level Distribution

Why this chart:

- Glucose level is a continuous number, so binning it into ranges is the right way to see the shape normal, skewed, or multiple peaks.

Interpretation

· Most patients fall between about 55 and 110 mg/dL, peaking around 80–90.

· A separate, smaller cluster of patients sits between roughly 180 and 240 mg/dL.

Storytelling:

The histogram of **Average Glucose Level** illustrates how blood glucose measurements are distributed across the study population. Because glucose level is a continuous numerical variable, a histogram effectively shows where most observations are concentrated and whether unusually high or low values exist. The summary statistics further support the visualization by describing the average glucose level, its spread, and the range of observed values.The chart tells the story of a population in which “most individuals have glucose levels within the lower to moderate range”, while a smaller number of participants exhibit considerably higher glucose levels. This creates a “positively skewed (right-skewed) distribution”, where the majority of observations are clustered at lower values, and a long tail extends toward higher glucose measurements.

This pattern is expected in real-world healthcare data because elevated blood glucose levels are typically confined to individuals with diabetes, prediabetes, or impaired glucose regulation rather than occurring throughout the entire population. Consequently, only a relatively small proportion of participants contribute to the higher end of the distribution.From a medical perspective, elevated blood glucose is one of the most significant risk factors for stroke. Persistently high glucose levels damage blood vessels, accelerate atherosclerosis, and increase the likelihood of blood clot formation, all of which contribute to cerebrovascular disease. Numerous studies have demonstrated that individuals with diabetes have approximately “twice the risk of experiencing a stroke” compared with individuals without diabetes (American Diabetes Association, 2024).

The histogram may also contain several “extreme high glucose values”, which appear as potential outliers. These values should not automatically be removed because they may represent patients with poorly controlled diabetes or acute hyperglycaemia. In medical datasets, such observations often contain valuable clinical information rather than data entry errors.From a data analytics perspective, the positive skewness observed in glucose levels suggests that the variable is not normally distributed. This has important implications for statistical analysis and predictive modelling, as certain algorithms perform better after data transformation or standardization. Nevertheless, retaining the original distribution is often appropriate because elevated glucose levels are themselves highly informative predictors of stroke risk.

Overall, the glucose distribution emphasizes that while most participants maintain relatively normal glucose levels, a smaller high-risk subgroup exhibits substantially elevated values. Identifying these individuals is crucial because early diagnosis, lifestyle modification, and effective diabetes management can significantly reduce the risk of stroke and other cardiovascular complications (World Health Organization, 2025).

Notable Observations

· The distribution is bimodal — there are two distinct groups: one with normal glucose, one much higher, likely diabetic or pre-diabetic patients.

· Because of this skew, the average (mean) will look higher than what's typical for most people, so the median is a more honest summary.

4. BMI Distribution

Chart Type: Boxplot

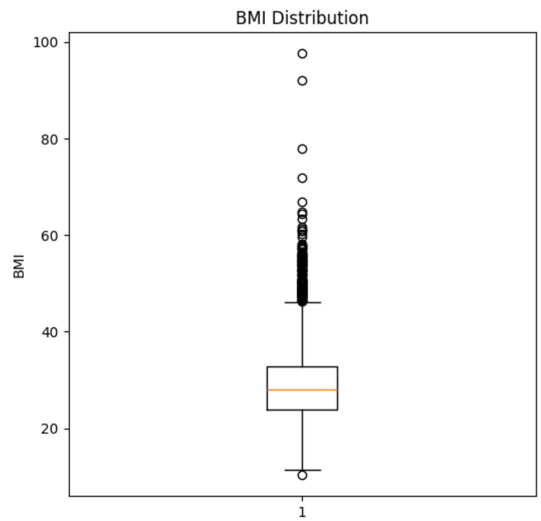


Figure — BMI Distribution

Why this chart:

- BMI is continuous, but the goal isn't just the shape, it's to clearly show the median, the middle 50% of values, and flag outliers.
- A box plot does all three in one compact picture, which a histogram doesn't.

Interpretation

· The median BMI is around 28.

· Half of all patients fall between about 24 and 33 (the box).

· A long stack of points from about 46 up to 90+ are flagged as outliers.

Storytelling:

The "Body Mass Index (BMI)" histogram provides a clear picture of the weight status of individuals in the dataset. Unlike categorical variables, BMI is a continuous measurement, making a histogram the most appropriate visualization for understanding how participants are distributed across different BMI values. The accompanying summary statistics—including the mean, median, minimum, maximum, and quartiles—offer additional insight into the central tendency and variability of BMI within the population.The chart tells the story of a population with diverse body weight characteristics. Most participants are likely clustered around the **normal weight to overweight BMI range**, while progressively fewer individuals appear at the extreme low and high ends of the distribution. This pattern is commonly observed in population health studies, where the majority of adults fall within moderate BMI categories, with smaller proportions classified as underweight or obese.

From a clinical perspective, BMI is an important indicator because excessive body weight is associated with several conditions that increase stroke risk, including hypertension, type 2 diabetes, elevated cholesterol, and cardiovascular disease. Individuals with obesity often experience chronic inflammation and impaired blood vessel function, which can contribute to the formation of blood clots and increase the likelihood of stroke (World Health Organization, 2024). The histogram may also reveal a "slight positive (right) skew", where a relatively small number of individuals have exceptionally high BMI values. These observations represent individuals with severe obesity, who may require closer clinical attention because they are more susceptible to multiple chronic diseases. If the distribution contains unusually high BMI values, they may appear as potential outliers. Rather than treating these observations as errors, analysts should recognize that they may represent genuine patients with elevated health risks and should only remove them after careful clinical consideration.

From a data analytics standpoint, examining the distribution of BMI is essential before predictive modelling. If the histogram shows noticeable skewness or extreme values, transformations or robust statistical methods may be considered depending on the modelling technique. However, because high BMI is a clinically meaningful risk factor, preserving these observations is often preferable in medical datasets.Overall, the BMI distribution highlights that body weight varies considerably among participants and reinforces the importance of maintaining a healthy weight. Although BMI alone does not determine whether an individual will experience a stroke, it interacts with other risk factors such as age, hypertension, diabetes, and physical inactivity, making it a valuable predictor in stroke risk assessment (Centers for Disease Control and Prevention, 2024).

Notable Observations

· A median of 28 already falls in the "overweight" range, meaning many patients carry extra cardiovascular risk.

· Outliers are almost all on the high side, showing the skew leans toward high BMI, not low.

· The single point near 11 is almost certainly a data entry mistake, since that BMI isn't realistic.

· The cluster of outliers between 46–65 is large enough to be its own high-BMI subgroup, not just a few rare cases.

5.Hypertension Distribution

Chart Type: Bar Chart

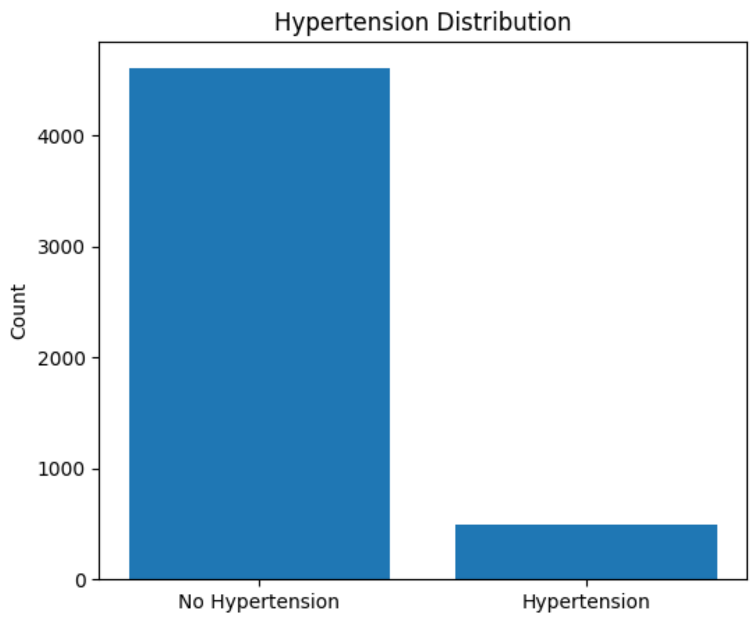


Figure — Hypertension Distribution

Why this chart:

- Hypertension is a simple yes/no category, so a bar chart makes comparing the two group sizes instant and clear.

Interpretation

· About 4,600 patients don't have hypertension.

· About 500 patients do.

Storytelling:

The “Hypertension Status” chart compares participants diagnosed with hypertension (“1”) against those without hypertension (“0”). The visualization shows that “most individuals in the dataset do not have hypertension, while a considerably smaller proportion have been diagnosed with the condition.This distribution tells the story of a population in which hypertension is present among a minority of participants, yet its significance extends far beyond its frequency. Although fewer individuals have hypertension, they represent one of the highest-risk groups for stroke. High blood pressure is widely recognized as the “single most important modifiable risk factor for stroke”, contributing to both ischemic and hemorrhagic strokes (World Health Organization, 2025).

The chart demonstrates that hypertension is not widespread across the entire dataset, which is expected in a general population sample. Most individuals maintain normal blood pressure, while a smaller subset experiences elevated blood pressure requiring medical attention. Despite being the minority, these individuals deserve particular focus because prolonged hypertension gradually damages blood vessels, weakens arterial walls, and promotes the formation of blood clots, significantly increasing the likelihood of stroke (Centers for Disease Control and Prevention, 2024).From a data analytics perspective, hypertension is recorded as a “binary categorical variable” (0 = No, 1 = Yes), making a bar chart the most appropriate visualization. Unlike continuous variables such as BMI or glucose level, binary variables represent distinct categories rather than numerical measurements. The chart provides a quick overview of class frequencies and allows analysts to identify whether the variable is highly imbalanced.

Although the majority of participants do not have hypertension, the chart alone does **not** indicate whether hypertensive individuals experienced more strokes. To investigate this relationship, a cross-tabulation or grouped visualization comparing **hypertension status** with the “stroke outcome” would be required. Such analyses often reveal that stroke prevalence is substantially higher among hypertensive patients than among individuals with normal blood pressure.From a predictive modelling perspective, hypertension is one of the most valuable features in stroke prediction models. Numerous clinical studies have consistently identified hypertension as a strong predictor of future stroke events because it directly contributes to vascular damage over time (Feigin et al., 2021). Consequently, this variable is expected to play an important role during feature selection and model development.

Overall, the hypertension distribution highlights a population in which most individuals have normal blood pressure, while a smaller but clinically significant group lives with hypertension. This reinforces the importance of routine blood pressure screening, early diagnosis, lifestyle modification, and appropriate treatment as essential strategies for reducing stroke risk and improving long-term cardiovascular health.

Notable Observations

· This is a 90/10 split a real imbalance.

· If hypertension is later used to help predict something like stroke, this imbalance has to be kept in mind, since a model could just assume "no hypertension" most of the time and still look accurate.

6.Heart Disease Distribution

Chart Type: Bar Chart

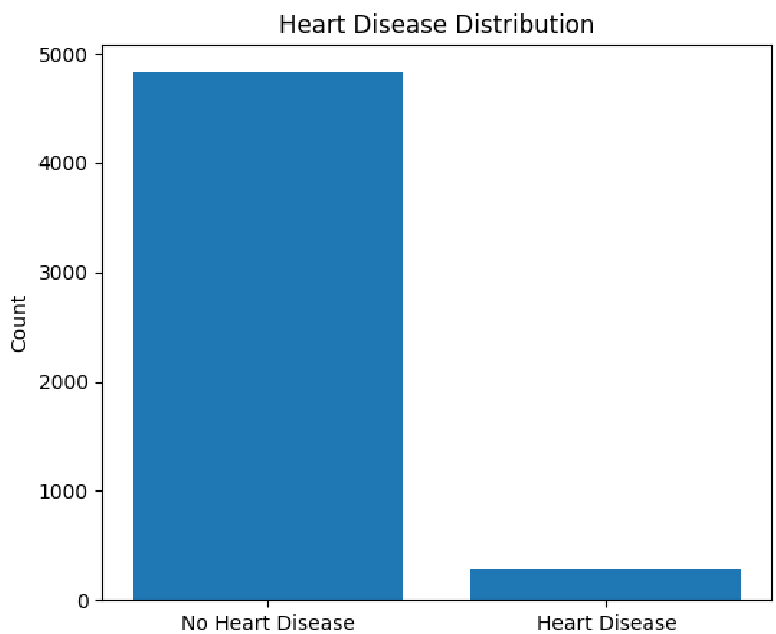


Figure — Heart Disease Distribution

Why this chart:

- Same reasoning as hypertension a yes/no category, so bars make the split easy to read.

Interpretation

· About 4,800 patients have no heart disease.

· Fewer than 300 patients do.

Notable Observations

· This imbalance (roughly 95/5) is even more extreme than hypertension.

· Despite being rare, heart disease could still be an important stroke warning sign and shouldn't be dropped just because it's a minority group.

Storytelling:

The “Heart Disease Status”chart presents the distribution of participants based on whether they have been diagnosed with heart disease (“1”) or not (“0”). The visualization shows that “the vast majority of individuals do not have heart disease”, while only a relatively small proportion have a documented history of the condition. This indicates that heart disease is less common within the study population, but its clinical importance should not be underestimated.The chart tells the story of a population in which most individuals appear to have healthy cardiovascular systems, yet a small subset carries one of the strongest predictors of adverse health outcomes. Although patients with heart disease represent the minority, they are among those most vulnerable to stroke because many cardiovascular conditions directly affect blood circulation to the brain.

From a medical perspective, heart disease is closely linked to stroke through several biological mechanisms. Conditions such as **coronary artery disease, atrial fibrillation, heart failure, and valvular heart disease** increase the likelihood of blood clot formation. These clots can travel through the bloodstream and block blood vessels supplying the brain, leading to an **ischemic stroke**, the most common type of stroke worldwide (World Health Organization, 2025). Consequently, even though the number of participants with heart disease is relatively small, they represent a clinically significant high-risk group.The distribution observed in the chart is typical of many population-based health datasets. Most people do not have diagnosed heart disease, while only a minority have existing cardiovascular conditions. This reflects the general prevalence of heart disease in the broader population and suggests that the dataset includes both healthy individuals and those with chronic cardiovascular illnesses, enabling meaningful comparisons between these groups.

From a data analytics perspective, **heart disease is a binary categorical variable**, coded as **0 (No)** and **1 (Yes)**. A bar chart is therefore the appropriate visualization because it effectively compares the frequencies of the two categories. The chart also allows analysts to assess whether the variable is highly imbalanced before incorporating it into statistical analyses or predictive models.Although participants with heart disease are fewer in number, the chart alone does **not** indicate that they experienced more strokes. To determine whether heart disease is associated with stroke occurrence, further analysis—such as comparing **heart disease status** with the **stroke outcome** using cross-tabulations or grouped visualizations—is required. Such analyses often reveal that stroke prevalence is considerably higher among individuals with heart disease than among those without cardiovascular conditions.

From a predictive modelling standpoint, heart disease is widely regarded as one of the most informative clinical features for stroke prediction. Numerous machine learning studies consistently identify heart disease as an important predictor because it captures underlying cardiovascular dysfunction that substantially elevates stroke risk. When combined with variables such as age, hypertension, average glucose level, BMI, and smoking status, heart disease helps improve the predictive performance of stroke risk models (Feigin et al., 2021).

Overall, the heart disease distribution chart highlights a population in which cardiovascular disease affects only a minority of participants, yet those individuals represent one of the highest-risk groups for stroke. The visualization reinforces the importance of early diagnosis, regular cardiovascular screening, effective treatment, and lifestyle modification in reducing stroke incidence. It also demonstrates why heart disease remains a key variable in both clinical decision-making and predictive health analytics.

7.Smoking Status Distribution

Chart Type: Bar Chart (Horizontal)

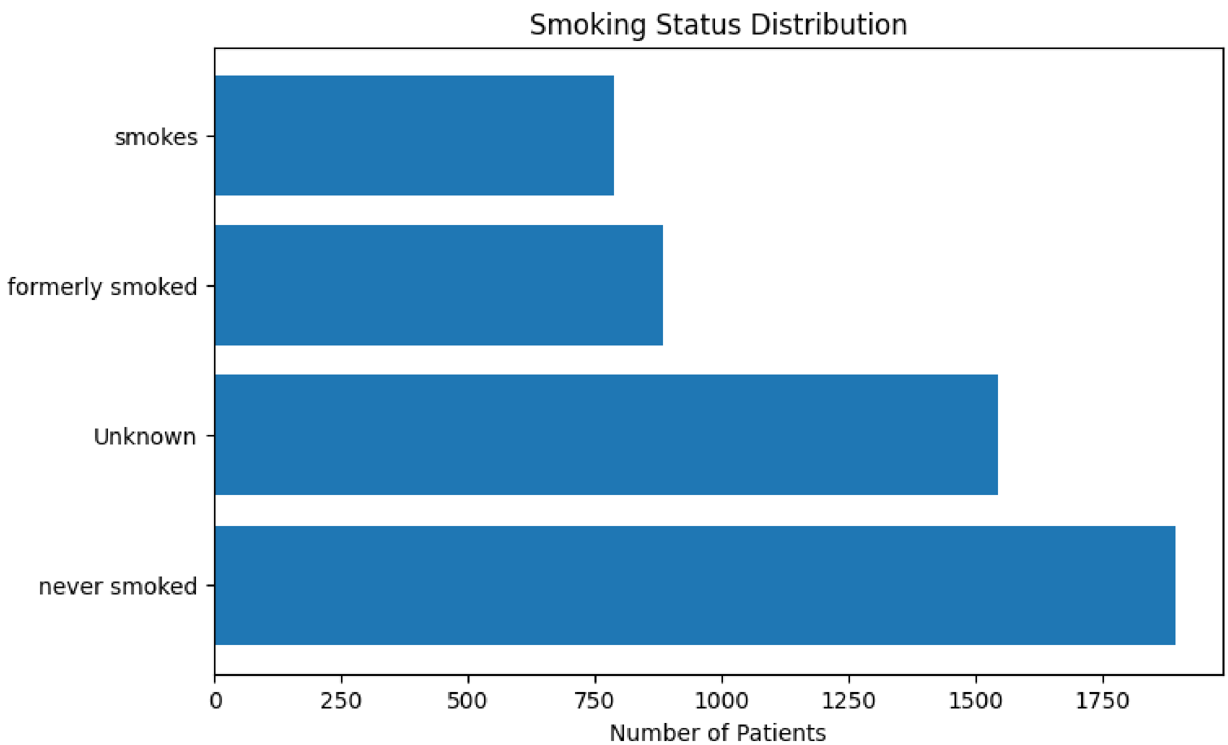


Figure — Smoking Status Distribution

Why this chart:

- Smoking status has four categories, some with long names ("formerly smoked," "never smoked").
- Turning the bars sideways gives the labels room to be read clearly without squeezing or rotating text

Interpretation

The horizontal bar chart breaks patients into four smoking categories. "Never smoked" is the largest group at nearly 1,900, followed by "Unknown" at around 1,550. "Formerly smoked" and "smokes" are smaller, at roughly 880 and 790 respectively.

Storytelling:

The smoking status distribution chart provides valuable insight into the lifestyle characteristics of the individuals represented in the dataset. The visualization categorizes participants into four groups: **formerly smoked**, **never smoked**, **smokes**, and **Unknown**. From the chart, it is evident that the **"never smoked" category contains the largest number of individuals**, followed by those with an **unknown smoking status**, while **current smokers** and **former smokers** make up smaller portions of the dataset.This distribution tells the story of a population in which the majority of participants report never having smoked. At first glance, this appears encouraging, as smoking is one of the most significant modifiable risk factors for stroke and other cardiovascular diseases. Individuals who avoid smoking generally have a lower risk of developing conditions such as atherosclerosis, hypertension, and blood vessel damage, all of which increase the likelihood of stroke (World Health Organization, 2025).

However, the chart also reveals a substantial number of participants with an **"Unknown"** smoking status. From a data analysis perspective, this is an important observation. The "Unknown" category may represent individuals who chose not to disclose their smoking habits, records with missing information, or data that could not be verified during collection. A large proportion of unknown values can limit the reliability of analyses involving smoking behavior because it introduces uncertainty and may mask the true relationship between smoking and stroke risk. Therefore, analysts should carefully consider how to handle this category during preprocessing, such as treating it as a separate class or investigating whether imputation is appropriate.Although **current smokers** represent a smaller proportion of the dataset, their presence remains clinically significant. Smoking damages blood vessels, promotes the formation of blood clots, increases blood pressure, and reduces oxygen delivery throughout the body. These physiological changes substantially increase the risk of both ischemic and hemorrhagic stroke. Research has shown that smokers have approximately **twice the risk of stroke** compared with non-smokers, making smoking cessation one of the most effective preventive strategies (Centers for Disease Control and Prevention, 2024).

The **formerly smoked** category also provides an important narrative. These individuals may have quit smoking due to increased health awareness, medical advice, or previous health complications. While former smokers may still carry some residual cardiovascular risk from past tobacco exposure, research consistently demonstrates that the risk of stroke declines after smoking cessation and continues to decrease over time. This highlights the long-term benefits of quitting smoking, even after years of tobacco use (U.S. Surgeon General, 2020).From a data analytics perspective, the smoking status distribution suggests that the dataset captures individuals with diverse lifestyle behaviors, making it suitable for investigating how smoking interacts with other variables such as age, hypertension, heart disease, and average glucose levels. However, because the distribution is uneven—particularly due to the large "Unknown" category—analysts should interpret results cautiously and consider appropriate preprocessing techniques before developing predictive models.

Overall, the smoking status chart reinforces the importance of lifestyle factors in stroke prevention. While the majority of participants report never smoking, the presence of current and former smokers highlights populations that may benefit from targeted public health interventions. At the same time, the sizeable unknown category underscores the need for complete and accurate health records to improve the quality of clinical research and predictive analytics.

Notable Observations

- The "Unknown" category is large enough (close to 30% of "never smoked") to meaningfully affect any analysis that relies on smoking status it cannot simply be ignored or dropped.
- Combined, former and current smokers make up a substantial share of patients, so smoking-related risk analysis still has a reasonably sized sample despite "never smoked" being the mode.

8.Stroke Distribution

Chart Type: Pie Chart

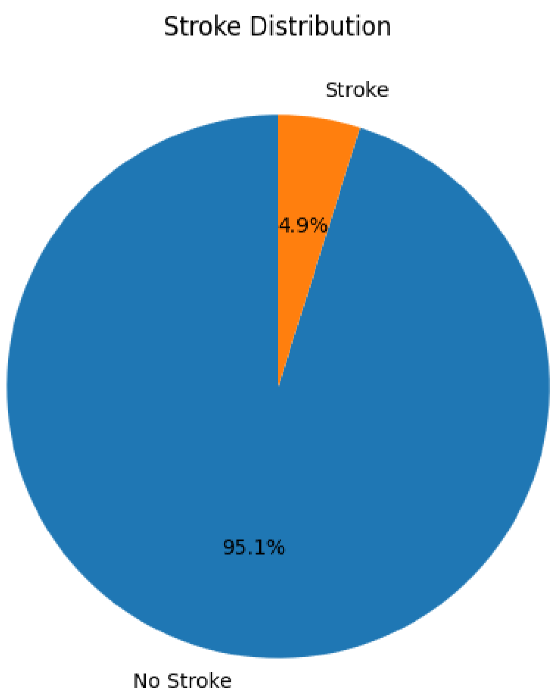


Figure — Stroke Distribution

Interpretation

The pie chart shows the target variable is heavily imbalanced: 95.1% of patients had no stroke, while only 4.9% experienced one.

Notable Observations

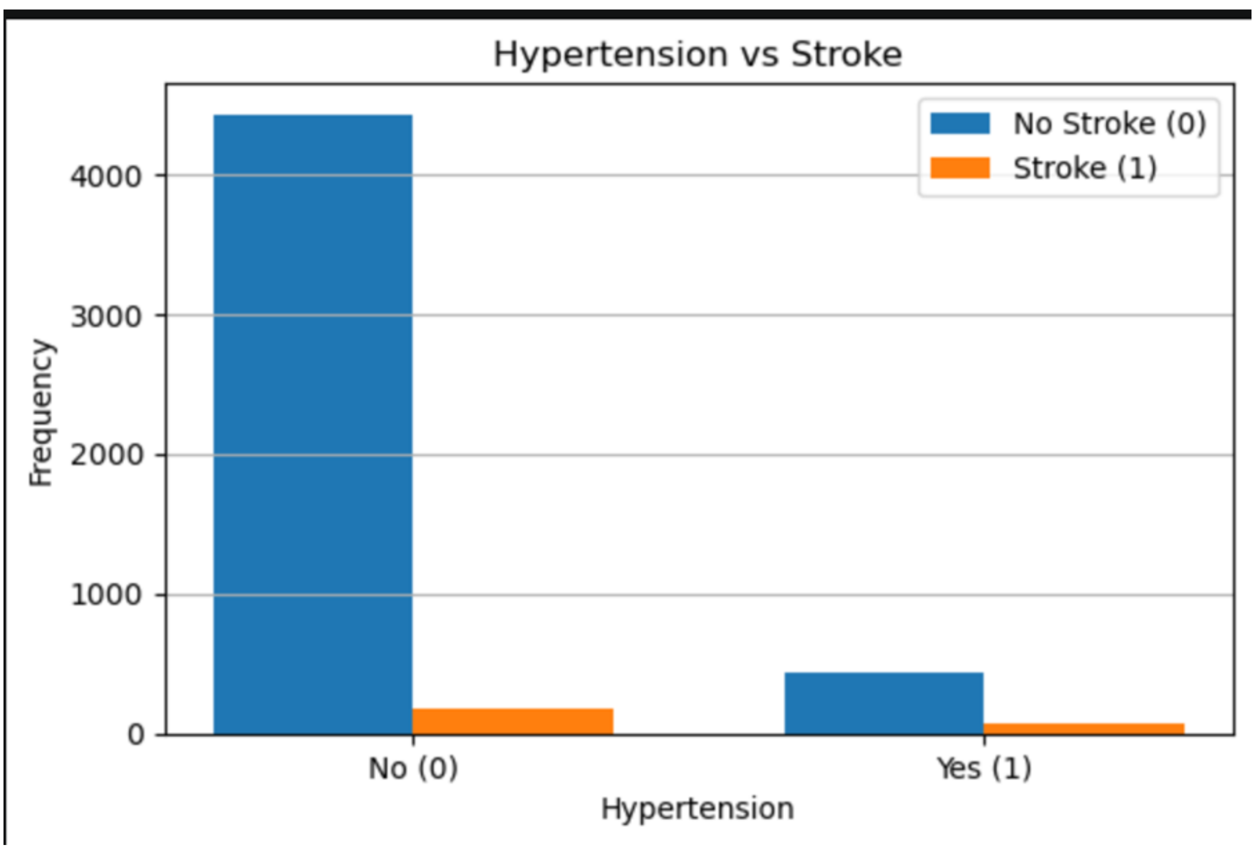
- This is a classic rare-event / class-imbalance problem. Any predictive model built on this target must account for the imbalance (e.g., resampling, class weighting, or appropriate evaluation metrics) or it risks simply predicting "no stroke" for everyone and still appearing highly accurate.
- Because stroke cases are scarce, larger sample sizes or techniques like SMOTE are typically needed to build a reliable classifier.

Overall Summary

Across all eight variables, two patterns stand out. First, several categorical health indicators hypertension, heart disease, and stroke itself are strongly imbalanced toward the "negative" class, which is expected for medical conditions but has real implications for downstream modeling. Second, the continuous variables (Age, Average Glucose Level, BMI) each show non-normal shapes: Age is fairly spread out with an elderly spike, while Glucose and BMI are both right-skewed with meaningful outlier populations. These characteristics justify the outlier handling and scaling steps applied earlier in the notebook, and they set expectations for the bivariate/multivariate analysis that follows.

3.4 Bivariate Analysis

1.Hypertwnsion vs Stroke – Grouped Bar Chart

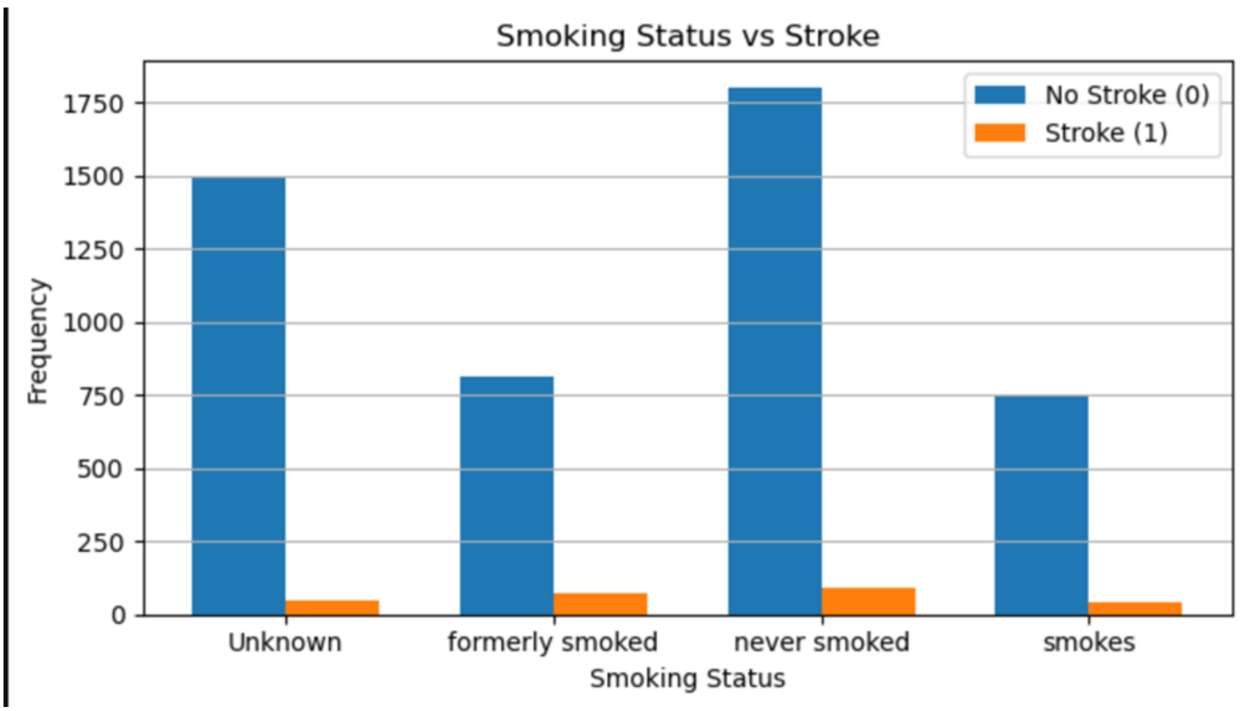


Interpretation: Similar to heart disease, hypertension acts as a severe clinical indicator. Patients with hypertension (Yes/1) exhibit a significantly larger ratio of stroke cases compared to the non-hypertensive group, validating the classical pathophysiological link between chronically elevated systemic arterial pressure and endothelial/cerebrovascular damage.

Storytelling

Stroke occurs much more frequently among participants with hypertension than among those without the condition. Although individuals without hypertension make up most of the dataset, the proportion of stroke cases is substantially higher among hypertensive patients. This confirms hypertension as one of the most important modifiable risk factors for stroke. The findings imply that improving blood pressure control could prevent many stroke cases.

2. Smoking vs Stroke---Grouped Bar Chart



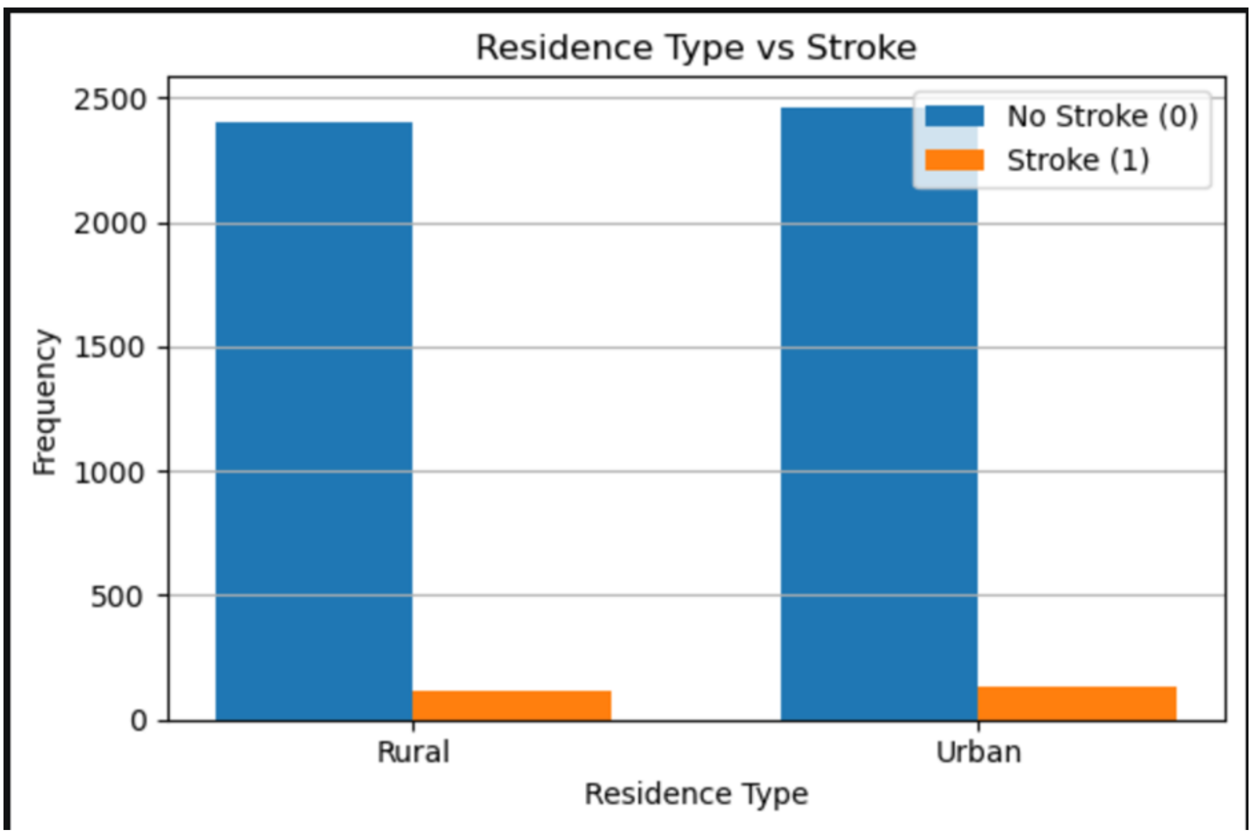
Interpretation: Individuals who 'formerly smoked' show a prominently visible proportion of stroke case relative to their sample size. The 'never smoked' group has the highest absolute count of strokes, but this is a reflection of its dominant status as the largest baseline cohort in the dataset.

Storytelling

Stroke cases are observed across all smoking categories, although former smokers appear to account for a relatively larger proportion of stroke cases than current smokers and individuals who never smoked. This pattern may reflect the fact that many former smokers are older adults or individuals who quit smoking following health complications. Nevertheless, the findings support extensive evidence that tobacco exposure increases stroke risk by damaging blood vessels and promoting atherosclerosis. Smoking cessation remains one of the most effective strategies for reducing stroke risk.

3. Residence Type vs Stroke

Chart Type: Grouped Bar Chart



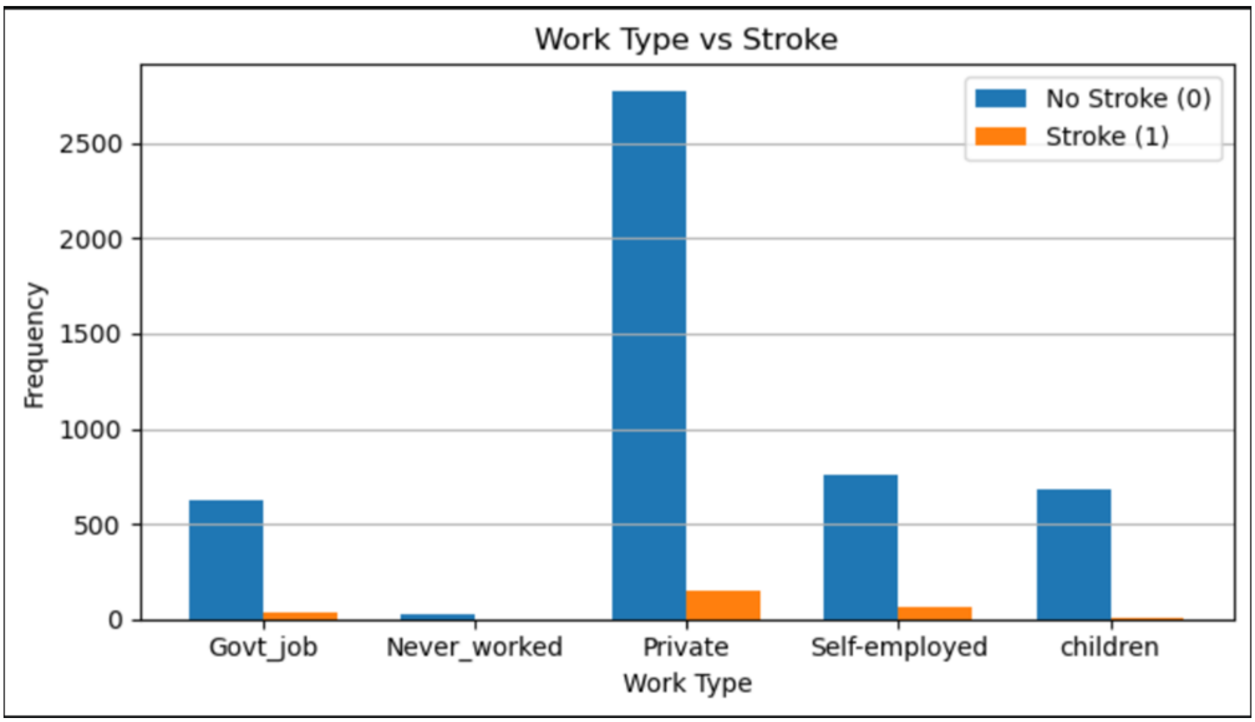
Interpretation: The distribution of both stroke events and non-stroke controls is almost perfectly symmetric between Rural and Urban populations. This suggests that geographic or residential location alone does not introduce a significant bias or standalone risk variance for stroke incidence within this dataset.

Storytelling

Stroke occurrence appears relatively similar between urban and rural residents, with no substantial differences observed. This suggests that residence type alone has limited influence on stroke risk within the dataset. Instead, medical conditions and lifestyle-related factors are likely to explain a much greater proportion of stroke occurrence than geographic location.

4.Work type vs stroke

Chart Type: Grouped Bar Chart



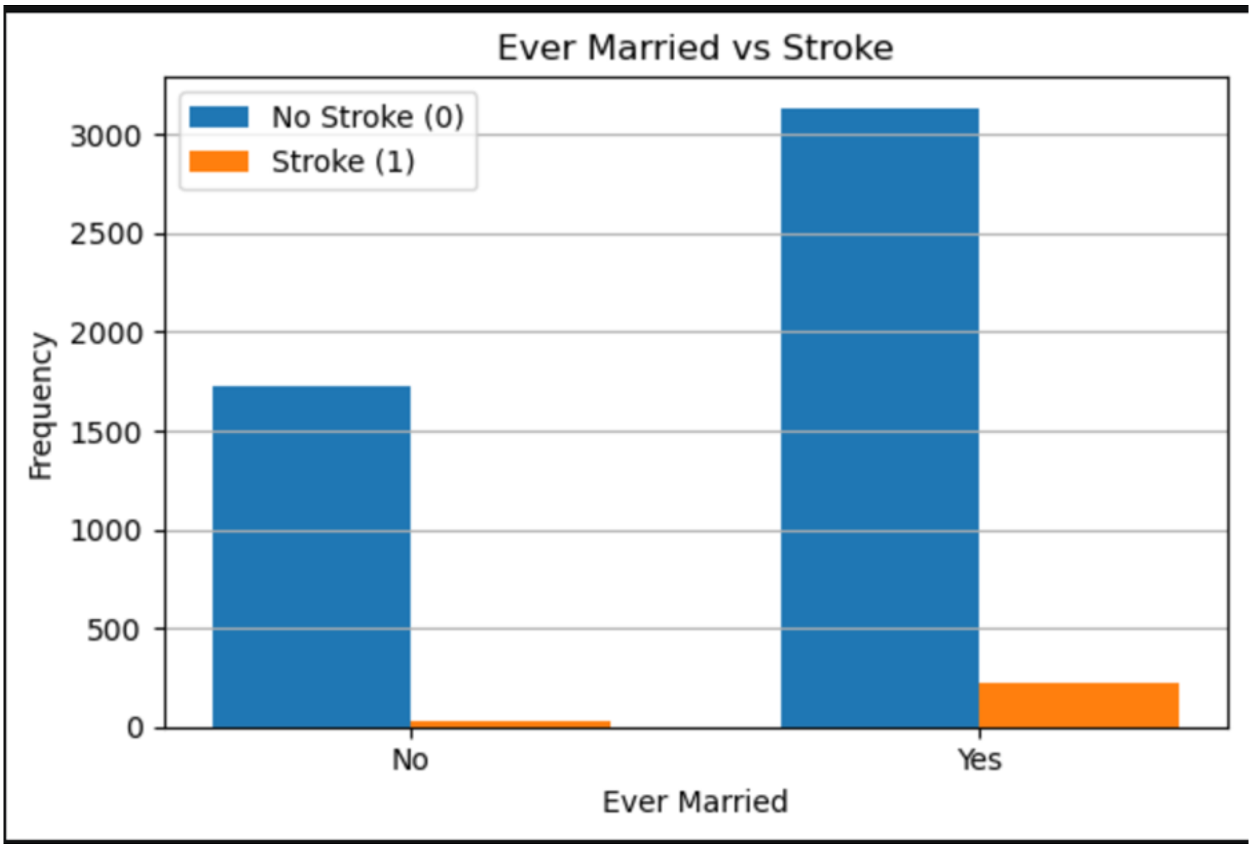
Interpretation: Stroke cases are predominantly found within the 'Private' and 'Self-employed' job sectors, while being virtually non-existent in children and individuals who have 'Never worked'. This reflects both the physical/stress-related toll of professional life and, crucially, the demographic distribution where children and younger individuals do not experience strokes at the same frequency as older, working-age or retired populations.

Storytelling

Stroke occurrence varies across different occupational categories, with self-employed and private-sector workers contributing a relatively larger share of stroke cases. However, these differences are likely influenced by age, employment patterns, and underlying health conditions rather than occupation itself. The findings imply that work type should be interpreted cautiously and analysed alongside demographic and clinical variables.

5.Ever Married vs Stroke

Chart Type: Grouped Bar Chart



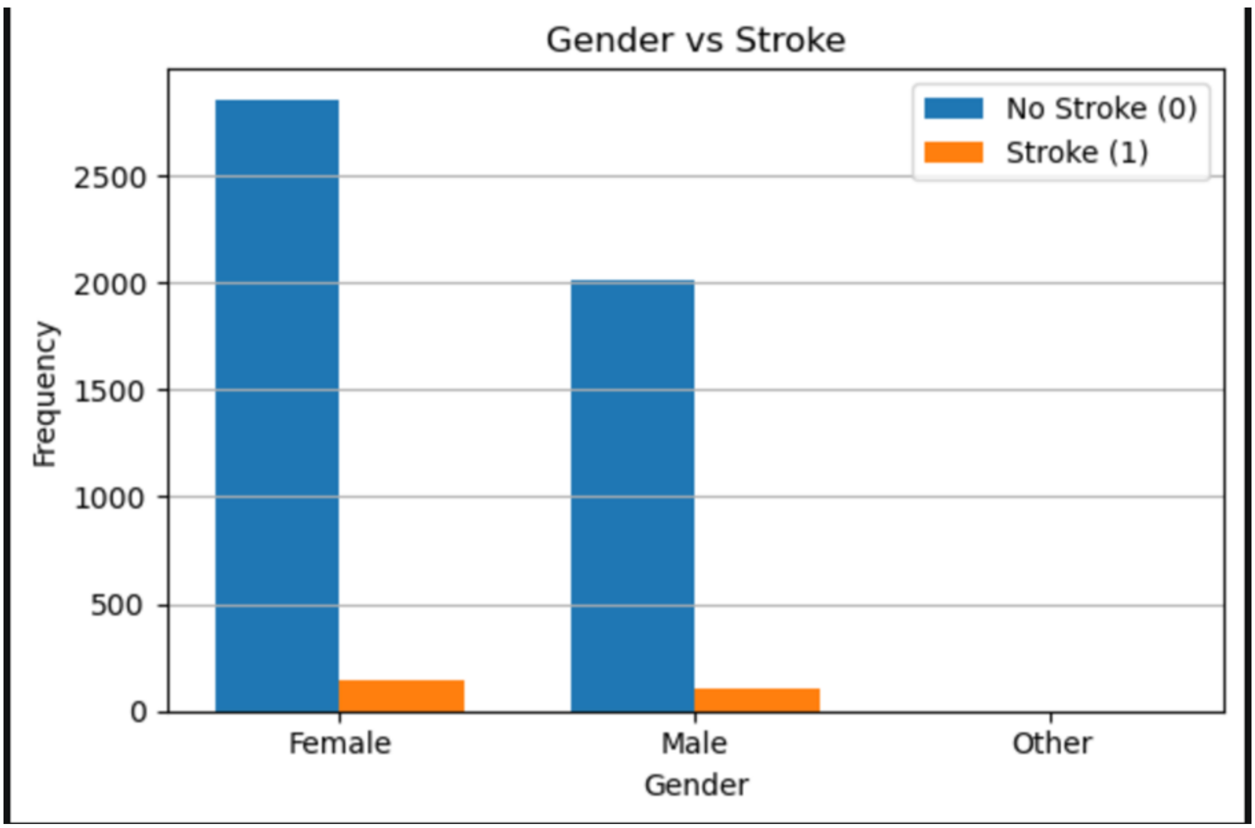
Interpretation: Individuals who have 'Ever Married' demonstrate a higher absolute and relative rate of stroke compared to those who have never been married. This association is primarily driven by a confounding variable: age. Married individuals in the dataset belong to a significantly older age bracket on average, which inherently elevates their stroke risk.

Storytelling

Participants who have been married account for more stroke cases than those who have never married. However, this relationship is most likely explained by age because married individuals tend to be older. Consequently, marital status should not be viewed as an independent cause of stroke. Instead, it serves as a demographic characteristic that may be associated with other underlying risk factors.

6.Gender vs Stroke

Chart Type: Grouped Bar Chart



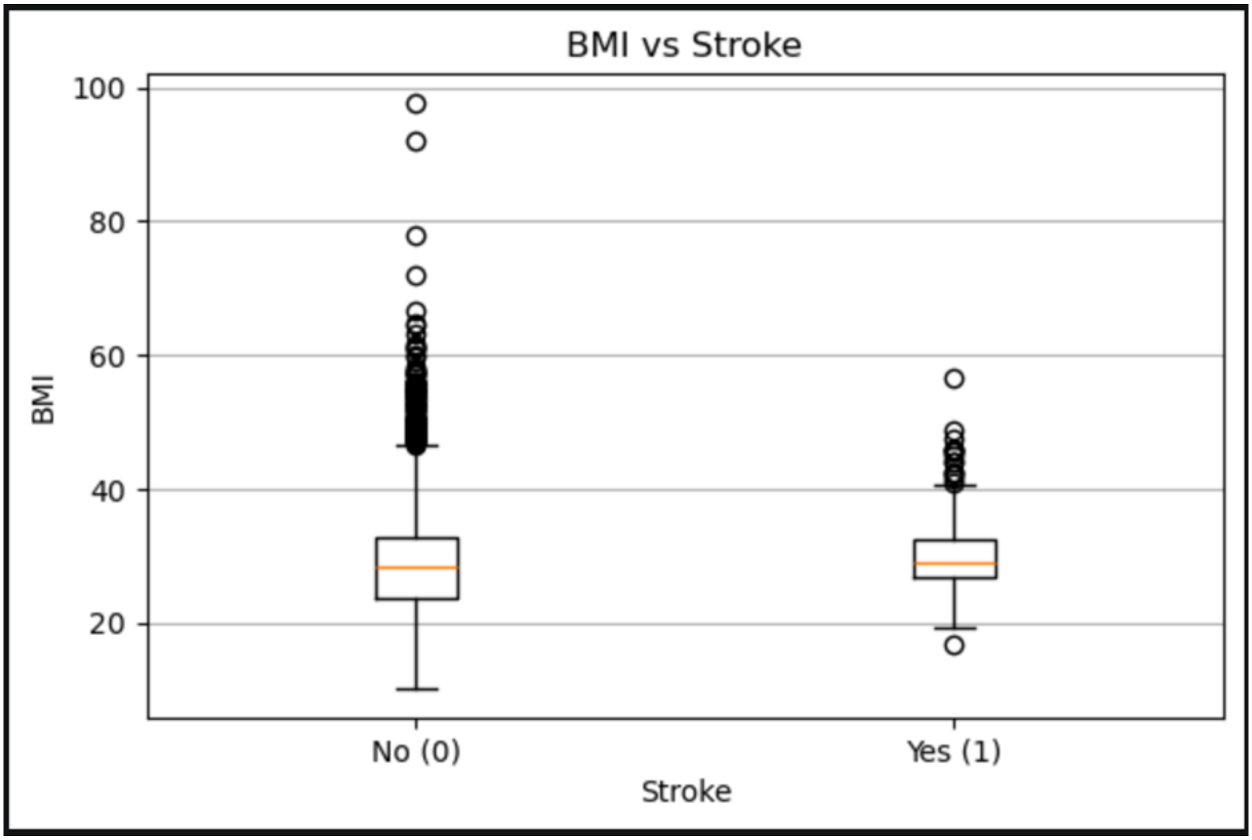
Interpretation: This distribution reveals that the dataset contains a larger baseline proportion of females than males. When evaluating the stroke rates within each category, both male and female groups exhibit highly similar relative frequencies of stroke events, indicating that gender may not carry as strong a standalone predictive weight compared to age or clinical comorbidities.

Storytelling

Stroke cases occur among both males and females, with only modest differences between the two groups. This suggests that gender alone is not a dominant predictor of stroke within this dataset. Instead, factors such as age, hypertension, diabetes, and heart disease appear to have a greater influence on stroke occurrence. Therefore, prevention strategies should focus primarily on clinical risk factors rather than gender alone.

7.BMI vs Stroke

Chart Type: Boxplot



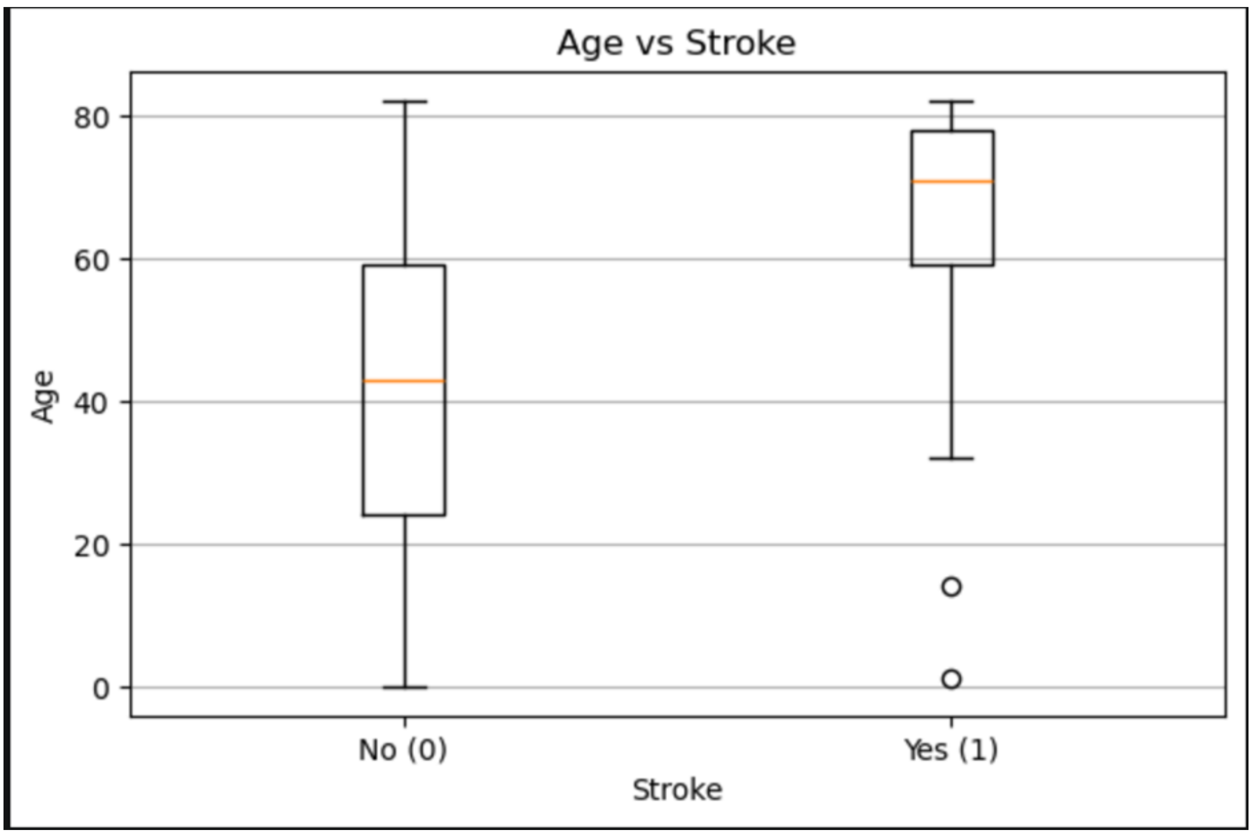
Interpretation: Interestingly, the median Body Mass Index (BMI) remains remarkably similar between the stroke and non-stroke groups (hovering around 28-29, which falls into the overweight category). However, the non-stroke group exhibits a massive number of extreme high-BMI outliers (reaching up to nearly 100), suggesting that while BMI is a known metabolic health factor, its direct linear relationship with stroke on a standalone basis is less pronounced than age.

Storytelling

The BMI distributions for stroke and non-stroke patients overlap considerably, although stroke patients exhibit a slightly higher median BMI. This indicates that BMI alone is not a strong independent predictor of stroke within this dataset. Instead, excess body weight may contribute indirectly by increasing the likelihood of hypertension, diabetes, and cardiovascular disease. These findings suggest that BMI should be considered alongside other clinical risk factors.

8.Age vs Stroke

Chart Type:Boxplot



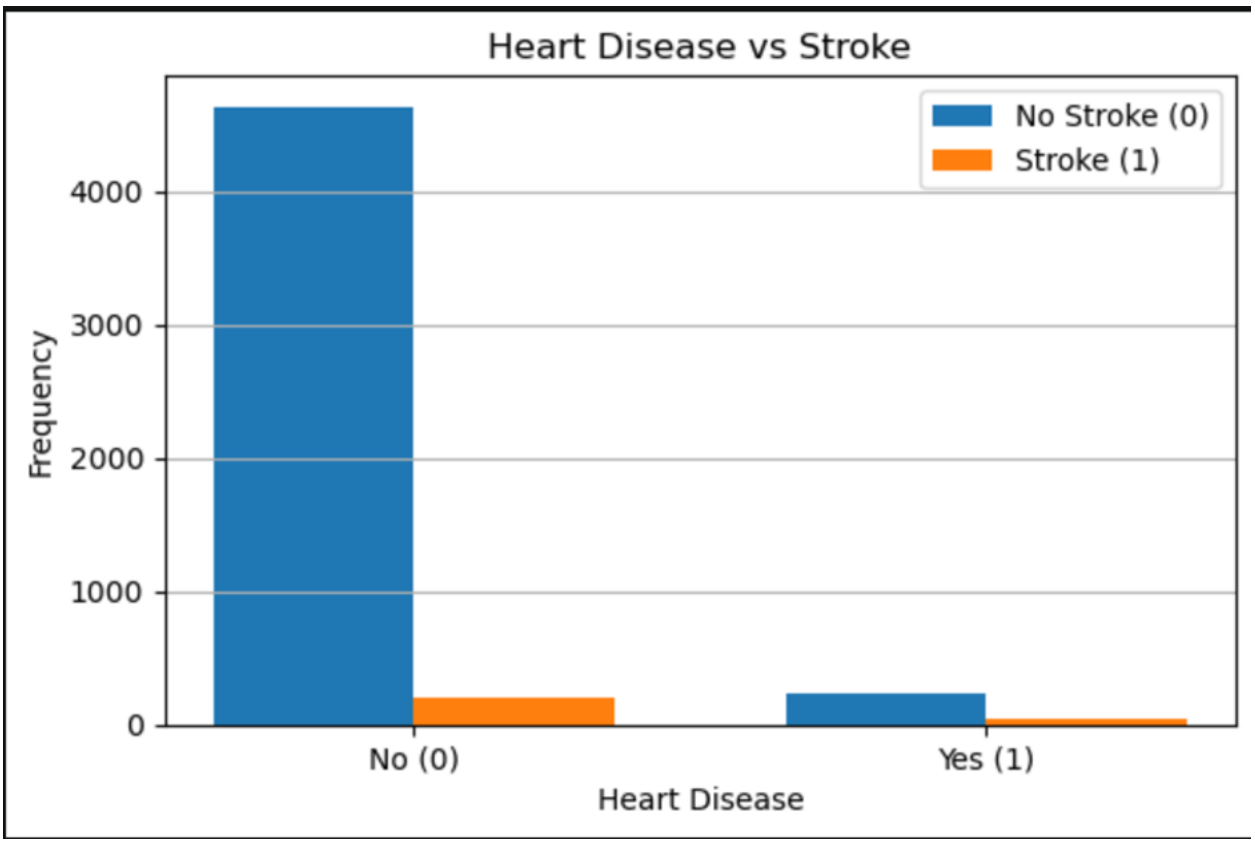
Interpretation: The boxplot illustrates a profound differentiation in age distributions between stroke and nonstroke patients. The median age for individuals who experienced a stroke is significantly higher (around 70+ years old) compared to the non-stroke group (median around 43 years old). The stroke distribution is heavily concentrated in older demographics, with a few low-age outliers, establishing age as one of the strongest continuous predictors of stroke incidence.

Storytelling

The boxplot demonstrates that individuals who experienced a stroke are generally older than those who did not. Stroke patients have a noticeably higher median age and are concentrated within the upper age ranges. This finding reinforces age as one of the strongest non-modifiable predictors of stroke. It implies that preventive healthcare initiatives should prioritise older adults through routine screening, blood pressure monitoring, and management of chronic diseases.

9.Heart Disease vs Stroke

Chart Type: Grouped Bar Chart



Interpretation: This grouped bar chart explores the relationship between pre-existing heart disease and stroke outcomes. While the absolute number of stroke cases is low across both cohorts due to data imbalance, the *proportion* of individuals who suffered a stroke is substantially higher among those with preexisting heart disease compared to those without. This highlights heart disease as a critical comorbid risk factor for cerebrovascular events.

Storytelling

Participants diagnosed with heart disease experience stroke at a much higher rate than those without heart disease. This relationship suggests that cardiovascular disorders contribute significantly to stroke development through impaired blood circulation and increased clot formation. The results reinforce the importance of cardiovascular disease prevention and treatment in reducing stroke burden.

4. Conclusion and Recommendations

4.1 Conclusion

This exploratory data analysis worked through 5,110 patient records containing 12 variables spanning demographic information (age, gender), lifestyle and social factors (work type, residence type, marital status, smoking status), and clinical indicators (hypertension, heart disease, BMI, average glucose level), with stroke occurrence as the target outcome. The workflow followed a structured EDA process: first checking data quality (no duplicate records were found), then assessing missingness (only BMI had gaps, with 201 missing values imputed using the mean), followed by univariate analysis of each variable individually, and finally bivariate analysis pairing each variable against the stroke target using the appropriate chart type — scatter plots for numerical-vs-numerical relationships, boxplots for numerical-vs-categorical comparisons, and grouped bar charts for categorical-vs-categorical comparisons. It's worth noting upfront that the dataset is heavily imbalanced, with only 249 patients (4.9%) having experienced a stroke compared to 4,861 (95.1%) who had not — this imbalance doesn't invalidate the patterns found, but it does mean any future predictive model built on this data will need to account for it (e.g., through resampling or class-weighting), since a model could otherwise achieve high accuracy just by predicting "no stroke" every time.

Key discoveries and important variables

1. Age emerged as the single most dominant factor in the entire analysis.Across every view of the data — the univariate histogram, the age-vs-stroke boxplot, and the age-vs-glucose scatter plot colored by stroke outcome — the pattern was unmistakable: patients who suffered a stroke were, on average, nearly three decades older than those who didn't (median age 71 vs. 43). This is also reflected in the correlation coefficient between age and stroke (~0.25), the strongest of any numerical variable tested. Clinically, this tracks with well-documented medical understanding — stroke risk rises sharply with age due to the cumulative effects of vascular aging, arterial stiffening, and the compounding of other risk factors over time.

2. Hypertension and heart disease stood out as the strongest clinical predictors after age. Even though both conditions were relatively rare in this sample (hypertension affected about 9.7% of patients, heart disease about 5.4%), patients with either condition had a dramatically higher proportional stroke rate than those without — roughly 13.3% vs. 4.0% for hypertension, and 17.0% vs. 4.4% for heart disease. These were the widest proportional gaps of any categorical variable examined, reinforcing that pre-existing cardiovascular conditions are powerful risk multipliers, consistent with established clinical literature on stroke etiology.

3. Average glucose level showed a real, if more moderate, relationship with stroke. The boxplot comparison showed stroke patients had a noticeably higher median glucose level and a wider spread toward elevated values, and the correlation with stroke (~0.13) was meaningfully positive, though weaker than age. This lines up with the known role of hyperglycemia and diabetes as stroke risk contributors, even if glucose alone isn't as strong a differentiator as age or the two binary clinical flags.

4. Several variables that looked important on the surface turned out to be weak or confounded once examined more closely.BMI, despite being a widely recognized general health risk factor, showed almost no linear relationship with stroke in this dataset (correlation ~0.04), and its boxplot distributions for stroke vs. non-stroke patients overlapped substantially. Gender also showed almost no difference in stroke proportion between Male and Female patients. Residence type (Urban vs. Rural) likewise showed near-identical stroke rates. These findings suggest that, at least in this dataset, these variables don't carry strong independent predictive value on their own.

5. Some apparent trends were likely artifacts of age rather than genuine independent effects.Ever-married patients, self-employed workers, and formerly-smoked patients all showed elevated stroke rates compared to their counterparts — but each of these groupings naturally skews older: people who have been married tend to have had more time to age, self-employment is more common among established, older adults rather than the very young, and "formerly smoked" implies enough life history to have started and then quit smoking. Without controlling for age, these categorical patterns risk being misread as causal when they may simply be proxies for the same underlying age effect already captured directly by the age variable.

Overall trend and interpretation

Taken together, the analysis paints a coherent and clinically plausible picture: stroke risk in this dataset is driven primarily by a small cluster of well-established medical risk factors — advancing age, hypertension, heart disease, and elevated glucose — while most demographic and lifestyle categorical variables (gender, residence, work type, marital status) contribute little independent signal and instead often reflect age differences between groups. This is reassuring from a domain-knowledge standpoint, since it means the patterns uncovered through EDA align with what's already understood about stroke risk in clinical practice, rather than surfacing unexpected or spurious relationships.

Implications for next steps

For any future predictive modeling, this suggests age, hypertension, heart disease, and average glucose level should be treated as priority features, while weaker variables like BMI, gender, and residence type may add less value unless combined with interaction terms or examined within age-stratified subgroups. Given the class imbalance noted earlier, techniques like SMOTE, class weighting, or stratified sampling would likely be necessary to build a model that performs well at actually identifying the minority stroke-positive cases, rather than simply optimizing for overall accuracy.

4.2 Recommendations

Overall stroke rate in this data-set is 4.9%, but risk is highly concentrated in specific, identifiable groups — meaning targeted action, not blanket intervention, will have the most impact.

1. Encourage Regular Health Screening

- **Prioritize screening from age 45 onward.** Stroke rate rises sharply with age: under 30 (0.1%), 45–60 (5.0%), 60–75 (9.7%), and over 75 (20.6%). Annual screening should be routine for patients 45+, and semi-annual for patients 60+.
- **Make blood pressure and glucose checks standard, not optional.** Patients with hypertension had a 13.3% stroke rate vs. 4.0% without it. Average glucose levels were also notably higher in stroke cases (132.5 vs. 104.8 mg/dL), reinforcing glucose testing as a core screening item.
- **Add cardiac screening for older adults.** Heart disease was linked to a 17.0% stroke rate vs. 4.2% without it — among the strongest single predictors in the data.
- **Don't skip BMI at checkups.** Stroke patients averaged a modestly higher BMI (30.5 vs. 28.8), supporting BMI as a routine, low-cost screening metric.

2. Promote Healthy Lifestyles

- **Target smoking cessation support, especially for former smokers.** Formerly smoked patients had the highest stroke rate by smoking status (7.9%), higher than current smokers (5.3%) or those who never smoked (4.8%) — likely reflecting cumulative long-term damage. Cessation campaigns should continue supporting patients well after they quit.
- **Support glucose and weight management programs.** Given the clear gap in average glucose and BMI between stroke and non-stroke patients, community programs on diet, physical activity, and diabetes prevention directly address two of the data-set's clearest risk markers.
- **Frame lifestyle messaging around blood pressure control,** since hypertension shows one of the largest risk gaps in the data and is the most modifiable of the major risk factors.

3. Monitor High-Risk Patients

- **Flag patients with combined hypertension + heart disease for close monitoring.** These patients show the highest stroke rate in the data-set — 20.3%, roughly 6x the baseline — compared to 3.4% for patients with neither condition. This group should be prioritized for case management or follow-up outreach.
- **Build an age + co-morbidity risk tier.** Patients 60+ with hypertension and/or heart disease represent the data-set's highest-concentration risk group and warrant proactive check-ins rather than waiting for symptoms.
- **Track married and working-adult populations too.** Married patients showed a notably higher stroke rate (6.6% vs. 1.7% unmarried) and self-employed workers had the highest rate by occupation (7.9%) — likely reflecting older age skew, but still useful for outreach targeting.

Bottom Line

Age, hypertension, heart disease, and glucose/BMI levels are the data-set's strongest stroke signals — and they compound. A patient with two or more of these risk factors should move to a proactive monitoring track rather than standard annual care.

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